CS-233 ML Project – Milestone 1 Report

Gaming & Mental Health: Predicting Addiction Metrics

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1 Introduction

This report presents the implementation and evaluation of three baseline Machine Learning algorithms on the *Gaming and Mental Health Dataset*, a synthetic dataset of 2,000 downsampled users (1,600 train, 400 test) described by 13 features. We address two prediction tasks:

Regression: predicting a continuous addiction score (integer, 0–10).

Classification: predicting a categorical addiction level $\in \{0\text{: Low, }1\text{: Medium, }2\text{: High}\}$.

Three models were implemented: closed-form **Linear Regression** (regression), gradient-descent **Logistic Regression** (classification), and **K-Nearest Neighbors** (both tasks).

2 Methodology

2.1 Data Preparation

Two execution modes were implemented. In validation mode (default), the 1,600 training samples are split into an **80% training set** (1,280 samples) and a **20% validation set** (320 samples), keeping the 400 test samples strictly for final evaluation. In test mode (`-test`), the full 1,600 samples are used for training.

All continuous features are **z-score normalised** ($z = (x - \mu)/\sigma$), with μ and σ computed exclusively on training data to prevent data leakage. Normalisation is critical for Logistic Regression (gradient stability) and KNN (distance isotropy).

2.2 Model Implementations

Linear Regression solves the regression task via the closed-form pseudo-inverse: $\mathbf{w} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y}$, with a bias column appended to $\mathbf{X}$.

Logistic Regression handles multi-class classification ($K = 3$) using a Softmax framework optimised with full-batch gradient descent. Numerical stability is ensured by subtracting $\max(z)$ before the exponential. Learning rate lr and number of iterations are tunable hyperparameters.

KNN predicts labels based on Euclidean distance. For classification, it returns the majority class among the K nearest neighbours; for regression, it returns their mean value.

3 Experiments & Results

3.1 Hyperparameter Tuning

Hyperparameters were selected using **5-Fold Cross-Validation** on the training set. Figure 1 contains two graphs: the left graph shows the mean Accuracy and Macro F1-Score for KNN as a function of K, while the right graph shows the mean Accuracy and Macro F1-Score for Logistic Regression as a function of the learning rate lr, with fixed `max_iters`$= 1{,}000$.

For **KNN**, accuracy increases with K, while macro F1 is highest at K=1 and then recovers for larger K. We selected K=9 as a practical trade-off because it achieves near-maximum accuracy while maintaining a strong macro F1, outperforming intermediate choices like K=5 and K=7 on both metrics.

For **Logistic Regression**, performance remains flat for $lr \leq 10^{-3}$ (too slow to converge in 1,000 iterations) and peaks at $lr = 0.1$. At $lr = 1.0$, both metrics slightly drop, indicating the onset of gradient instability.

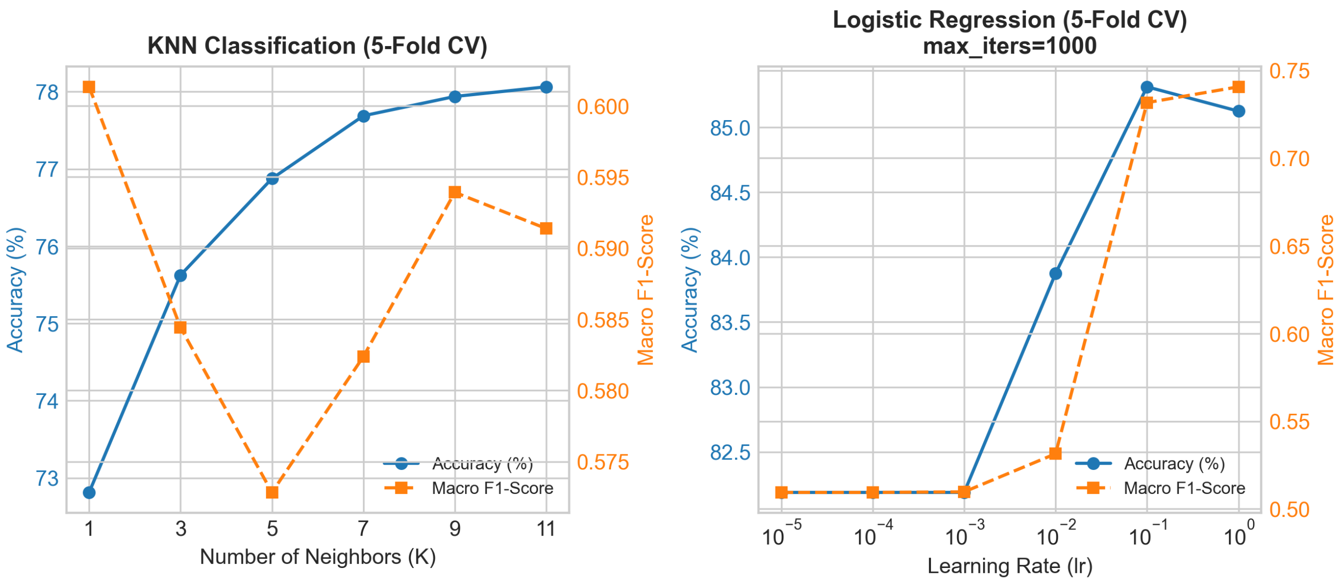

Figure 1: 5-Fold CV performance. *Left*: KNN classification accuracy and F1 vs. K. *Right*: Logistic Regression accuracy and F1 vs. learning rate.

3.2 Final Test Set Evaluation

All models were retrained on the full 1,600 training samples and evaluated on the 400 test samples. Results are reported in Table 1.

Table 1: Final test set performance.

Method	**HP**	**Acc.**	**F1**	**MSE**
Linear Reg.	–	–	–	**0.994**
KNN Reg.	K=9	–	–	1.865
Logistic Reg.	lr=0.1, it=1000	**88.25%**	**0.819**	–
KNN Classif.	K=9	79.75%	0.590	–

4 Discussion & Conclusion

Regression: Linear Regression substantially outperforms KNN (MSE 0.994 vs. 1.865). The addiction score likely follows a near-linear relationship with the features, which the closed-form solution captures exactly, while KNN's local averaging introduces bias.

Classification: Logistic Regression dominates KNN (88.25% vs. 79.75% accuracy; F1 0.819 vs. 0.590). The low KNN F1-score reveals its difficulty with class imbalance: it tends to over-predict the majority class, hurting per-class recall. Logistic Regression's gradient-optimised decision boundaries distribute predictions more uniformly across all three classes.

Runtime: Parametric methods (Linear, Logistic) are fast to predict (≈0s) after a one-time training cost (Logistic: ≈0.27s for 1,000 iterations). KNN has near-zero training time but slower inference (≈0.035s for 400 samples) due to the exhaustive distance computation over all 1,600 training points.

Conclusion: Properly tuned parametric models are clearly superior on this dataset, both in predictive performance and inference efficiency. Future work could explore regularisation for Logistic Regression or approximate nearest-neighbour methods to speed up KNN inference.